

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 07:00 UTC

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Generated 2026-05-10 07:00 UTC

- Cycles recorded: **526**
- Time span: 393.3h (~1.34 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	491
BARRIER_HIT	16
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	28
Free Probability	12
Schur-Weyl Duality	12
Representation Theory of Symmetric Groups	11
Matroid Theory	10
Algebraic Geometry	7
Random Matrix Theory	6
Additive Combinatorics	6
Finite Geometry	6
Spectral Graph Theory	6
Polymatroid Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Fourier analysis of boolean functions	3
Algebraic Topology	3

Field	Cycles
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Noncommutative L^p Geometry	3
Persistent Homology	3
Noncommutative Geometry	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-10 00:22 UTC	INCONCLUSIVE	Kronecker Coefficient Exponential Gap in Symmetric Powers of Perm
2026-05-10 00:30 UTC	INCONCLUSIVE	Operator Norm Lower Bound for Disjointness
2026-05-10 00:53 UTC	INCONCLUSIVE	Communication Matrices Tensor Rank Exponential Gap in Read-Twice vs Read-Once
2026-05-10 01:02 UTC	INCONCLUSIVE	Branching Projective Plane Line Count Bounds ABP Width for Monotone Circuit
2026-05-10 01:08 UTC	INCONCLUSIVE	Quantum Rank Lower Bounds for Disjointness
2026-05-10 01:23 UTC	INCONCLUSIVE	Communication Matrices Symmetric Tensor Rank Gap in Permanent vs Determinant
2026-05-10 01:47 UTC	INCONCLUSIVE	Decompositi Negative Eigenvalue Count Lower Bounds SOS Degree for Max-CUT
2026-05-10 03:39 UTC	INCONCLUSIVE	Plethysm Coefficient Gap in Homogeneous Polynomial Decompositions
2026-05-10 03:58 UTC	INCONCLUSIVE	Hypergraph Maximum Matching Inverse Proportional to ACC^0 Circuit
2026-05-10 04:08 UTC	INCONCLUSIVE	Finite Field Rank and Branching Program Width for Boolean Functio
2026-05-10 04:25 UTC	INCONCLUSIVE	Matroid Rank Inverse Proportional to ACC^0 Circuit Size for $GF(2)$

Time	Verdict	Title
2026-05-10 04:55 UTC	INCONCLUSIVE	Secant Variety Dimension Lower Bounds Disjointness Communication
2026-05-10 05:06 UTC	INCONCLUSIVE	Real Critical Point Count Exponential in SOS Degree for Max-CUT
2026-05-10 05:57 UTC	INCONCLUSIVE	Schatten p-Norm Lower Bound for Disjointness Communication Comple
2026-05-10 06:28 UTC	INCONCLUSIVE	SOS Degree Lower Bound via Adjacency Eigenvalue Multiplicity

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```